William Brown

Email: williamlandisbrown@gmail.com Website: www.astrotino.com Portfolio: www.astrotino.com/portfolio

Relevant Skills

Engineering: SolidWorks (CSWA), Flotherm XT, OpenFOAM, KiCAD,Visual Studio, Microsoft Office Suite
Programming: Python, Flutter, Kivy (Python), C++ (Arduino), Java, Git, Github, and Linux
Hands on: CNC, milling, 3D printing, knife plotter, soldering, PCB design, stepper motors

Engineering Experience

Northeastern University (2019), Boston, MA
Bachelor of Science in Mechanical Engineering with a Minor in Computer Science
NSF I-Corp Regional Participant 2023, Keller Innovation Forum Pitch competition

FLX Solutions, Inc., Knoxville, TN (remote) *April 2025 - Present*
Mechatronics Lead

- Worked with fellow engineers to iterate on and improve mechanical and electrical design of robot accessories — including PCB design, mechanism design, DFM, and DFA.
- Owned hardware build and delivery of inspection robot accessories; fabricated poles, handles, bases, and LED light assemblies. Supporting 8 robots deployed in the field, including one at every branch of the US Military.
- Led internal research efforts evaluating new sensing modalities and harsh-environment applications to expand platform capabilities.
- Onboarded and managed interns, including weekly check-ins, parts procurement, and grant milestone coordination.

Consonant Systems LLC, Philadelphia, PA (remote) *April 2023 - July 2025*
R&D Engineering Consultant

- Designed and optimized a fluorescence-based system using off the shelf evaluation boards, custom components, and custom code. Collaborated with the chemistry team to define device specifications and assess system performance. Developed and executed test protocols to evaluate the performance of the diagnostic system.
- Designed and implemented a device interface app in Flutter. Worked with a design consultant to improve the usability and quality of the UI.
- Developed a CFD model in OpenFOAM to support design work for microgravity applications.
- Participated as an Entrepreneurial Lead in the I-CORPs Regional Program at Princeton University
- Presented in the Keller Innovation Forum Pitch competition on behalf of a client company, placed 3rd winning a $10,000 prize to further aid in R&D.
- Manufactured over 1,000 laminated microfluidic devices for use in Exosome based liquid biopsy research.
- Prototyped a data acquisition and control system for a microfluidics client using Python, arduino, and ESP-32 dev boards.

Chip Diagnostics, Philadelphia, PA *September 2021 - April 2023*
Engineer

- Maintained and tested a legacy customer-facing GUI application written in the Python Kivy framework.
- Worked with customers in both internal and external labs to provide customer support and improve the workflow of our automated bench-top device. For example, adding a new section to the app where users could change the current protocol and a summary screen listing all steps of the selected protocol.
- Prototyped a next generation pressure drive system using Raspberry pi and Arduino modules for vacuum pump control.
- Brought production in house from an engineering consultancy including shop development and creation of an inventory management solution from the ground up. Re-designs and process improvements on the flagship product saved $5k per device. Roughly 50% of the original advertised sale price.
- Worked to develop and improve an in house production line for laminated microfluidic medical devices. Work included but was not limited to: design work, continual process improvements, materials planning, QA, and manual assembly.

Starry Inc, Boston, MA
Mechanical Design Co-Op, Mechanical Engineer 1 January - August 2018, June 2019 - August 2021

- Wrote and tested microcontroller software communicating over SPI using Python on a Raspberry Pi to unblock work on a mission critical company project
- Designed and managed the system level product tester for the company's newest and highest volume radio operating at a new frequency. Work included: system specification, RF schematic reviews, component design in SolidWorks, coordinating and ordering from vendors, manual part re-works in house, and final assembly
- Collaborated with RF engineers and systems engineers to help with various Python programming tasks to develop systems, collect, and inspect data of RF performance (EIRP, MCS level, and data-rate chief among them)

- Worked as a thermal analyst across multiple products. Ran thermal simulations using a combination Flotherm XT and Excel spreadsheets to validate and iterate on thermal designs quickly. One such project saved $330,000 dollars per one thousand radios built.

Background and Interests

- Eagle Scout (2/4/2014): Designed and constructed risers for a local middle school for my Eagle project
- Personal projects that explored neural networks and ROS2 using the Nvidia Jetson Nano platform